CH 4 — Linear Approximation

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4.1 Partial Differentiation

The "slope" of the surface depends on the direction of travel

> 📘 **Definition — Partial Differentiation**
>
> The **partial derivatives** of $f(x,y)$ are defined as
>
> $$\frac{\partial f}{\partial x} = \lim_{h\to 0} \frac{f(x+h,y)-f(x,y)}{h}$$
>
> $$\frac{\partial f}{\partial y} = \lim_{h\to 0} \frac{f(x,y+h)-f(x,y)}{h}$$

Examples:

1. $f(x,y) = x^3 \sin(xy)$

   $\frac{\partial f}{\partial x} = 3x^2 \sin xy + x^3 y \cos(xy)$

   $\frac{\partial f}{\partial y} = x^4 \cos(xy)$

Note: if the derivative rules do not apply, then use definition above.

2. $f(x,y) = (x^3+y^3)^{\frac{1}{3}}$, what is $\frac{\partial f(0,0)}{\partial x}$

   $\frac{\partial f}{\partial x} = x^2(x^3+y^3)^{-\frac{2}{3}}; x^2+y^2 \neq 0$

   $\lim_{h\to 0} \frac{f(x+h,y)-f(x,y)}{h} = \lim_{h\to 0} \frac{f(0+h)-f(0,0)}{h} = \lim_{h\to 0} \frac{(h^3)^{\frac{1}{3}}-0}{h} = 1$

3. Show that the above function where $\frac{\partial f}{\partial x}$ is not continuous at $(0,0)$

   $\lim_{(x,y)\to(0,0)} \frac{\partial f}{\partial x} = \lim_{(x,y)\to(0,0)} \frac{x^2}{(x^3+y^3)^{\frac{2}{3}}}$

   Consider path $y = mx$

   $\lim_{x\to 0} \frac{x}{(x^3+(mx)^3)^{\frac{2}{3}}} = \lim_{x\to 0} \frac{1}{(1+m^3)^{\frac{2}{3}}} \neq 1$

   The limit does not exist, so $\frac{\partial f}{\partial x}$ is not continuous at $(0,0)$

4. $f(x,y) = \begin{cases} \frac{2xy}{x^2+y^2} & (x,y)\neq(0,0) \\ 0 & (x,y)=(0,0) \end{cases}$

   $\frac{\partial f(0,0)}{\partial x} = \lim_{h\to 0} \frac{f(h,0)-f(0,0)}{h} = \lim_{h\to 0} \frac{\frac{2h\cdot 0}{h^2+0^2}-0}{0} = \lim_{h\to 0} \frac{0}{h} = 0$

   $\frac{\partial f(0,0)}{\partial y} = \lim_{h\to 0} \frac{f(0,h)-f(0,0)}{h} = \lim_{h\to 0} \frac{\left(2\frac{h}{h^2+0^2}\right)-0}{h} = \lim_{h\to 0} \frac{0}{h} = 0$

   Notice that this function is not continuous at $(0,0)$ but both partial derivatives exists at $(0,0)$

4.2 Higher-Order Partial Differentiation

Example:

Determine first and second derivative of $f(x, y) = x^2 y$

$$\frac{\partial f}{\partial x} = 2xy, \frac{\partial f}{\partial y} = x^2$$

$$\frac{\partial^2 f}{\partial x \partial x} = 2y, \frac{\partial^2 f}{\partial y \partial x} = 2x$$

$$\frac{\partial^2 f}{\partial x \partial y} = 2x, \frac{\partial^2 f}{\partial y \partial y} = 0$$

Short-hand notations

Leibniz Notation	Short-Hand Notation	Operator Notation
$\frac{\partial f}{\partial x}$	f_x	$D_1 f$
$\frac{\partial^2 f}{\partial x \partial x}$	f_{xx}	$D_1^2 f_x$
$\frac{\partial^2 f}{\partial x \partial y}$	f_{xy}	$D_2 D_1 f_x$
$\frac{\partial^2 f}{\partial y \partial x}$	f_{yx}	$D_1 D_2 f_x$
$\frac{\partial^2 f}{\partial y \partial y}$	f_{yy}	$D_2^2 f_x$

📐 **Theorem — Clairaut's Theorem**

If f_{xy} and f_{yx} are defined in some neighbourhood of (a, b) and are continuous at (a, b)

$$f_{xy}(a, b) = f_{yx}(a, b)$$

This means the most functions the "mixed" partial derivatives will be equal. A generalized of Clairaut's Theorem applies to higher-order partial derivatives.

Example

$f(x, y, z) = x^2 e^{y+z} \sin z$

$\frac{\partial f}{\partial x} = 2xe^{y+z} \sin z$

$\frac{\partial f}{\partial y} = x^2 e^{y+z} \sin z$

$\frac{\partial f}{\partial z} = x^2(e^{y+z} \sin z + e^{y+z} \cos z) = x^2 e^{y+z}(\sin z + \cos z)$

4.3 Tangent Plane

📘 **Definition — Tangent Plane**

The **tangent plane** to $z = f(x, y)$ at point $P = (a, b, f(a, b))$ is

$$z = f(a, b) + \frac{\partial f}{\partial x}(a, b)(x - a) + \frac{\partial f}{\partial y}(a, b)(y - b)$$

That is, a **tangent plane** is the plane which contains the tangent lines to all curves on $z = f(x, y)$ passing through point $(a, b, f(a, b))$

In particular, if

$C_1 = \text{cross-section corresponding to } y - b, z = f(a, b) + \frac{\partial f}{\partial x}(a, b)(x - a)$

$L_1 = \text{tangent line to } C_1 \text{ to } P$

$C_2 = \text{cross-section corresponding to } x - a, z = f(a, b) + \frac{\partial f}{\partial y}(a, b)(y - b)$

$L_2 = \text{tangent line to } C_2 \text{ to } P$

Example:

Let $z = \ln(2x + y)$, find tangent plane at $(-1, 3)$

$f(x, y) = \ln(2x + y), (a, b) = (-1, 3)$

$f(-1, 3) + \frac{\partial f}{\partial x}(-1, 3)(x + 1) + \frac{\partial f}{\partial y}(-1, 3)(y - 3) = 0 + 2\frac{x+1}{2(-1)+3} + \frac{y-3}{2(-1)+3} = 2(x + 1) + y - 3$

$= 2x + y - 1$

4.4 Linear Approximation

📘 **Definition — Linearization and Linear Approximation**

The **linearization** of f at (a, b) is the funciton

$$L_{(a,b)}(x, y) = f_x(a, b)(x - a) + f_y(a, b)(y - b) + f(a, b)$$

The **linearization** is a function.

The tangent plane is when we have $z = L_{(a,b)}(x, y)$ as a surface

Linear Approximation is when (x, y) is close to (a, b) we can say

$$f(x, y) \approx L_{(a,b)}(x, y)$$

Example:

Esimate $\sqrt{(0.95)^3 + (1.98)^3}$

Let $f(x, y) = \sqrt{x^3 + y^3} \implies f(0.95, 1.98) = \sqrt{(0.95)^3 + (1.98)^3}$

$\approx L_{(1,2)}(0.95, 1.98) = \frac{3(1)^2}{2((1)^3+(2)^3)^{\frac{1}{2}}}(x - 1) + \frac{3(2)^2}{2((1)^3+(2)^3)^{\frac{1}{2}}}(y - 2) + \sqrt{1^3 + 2^3}$

$= \frac{1}{2}(x - 1) + 2(y - 2) + 3 = \frac{1}{2}(-0.05) + 2(-0.02) + 3 = 2.935 \approx \sqrt{(0.95)^3 + (1.98)^3}$

📘 **Definition — Increment for of Linear Approximation**

Often we are itnerested in estiamting the **change** in some quantity. We can rewrite the lienar approximation in **increment form**:

$$\Delta f \approx f_x(a,b)\Delta x + f_y(a,b)\Delta y$$

for $\Delta f = f(x,y) - f(a,b), \Delta x = x - a, \Delta y = y - b$

Example:

Determine the volume of a cone $V(r,h) = \frac{\pi r^2 h}{3}$

$\Delta V \approx V_r(r,h)\Delta r + V_h(r,h)\Delta h = \frac{2\pi rh}{3}\Delta r + \frac{\pi r^2}{3}\Delta h$

Consider $r_i = 2, h_i = 10$, r increases by 0.1, h increases by 0.2

$\Delta V \approx \frac{4\pi}{3}(1) + \frac{4\pi}{3}(0.2) = \frac{8\pi}{5}\text{cm}^3$

4.5 Linear Approximation with Higher Dimensions

Let $\vec{x} = (x,y,z), \vec{a} = (a,b,c)$. We define the linearization of $f(\vec{x})$ at $\vec{a}$ as

$$\begin{aligned} L_{\vec{a}}(\vec{x}) &= f_x(\vec{a})(x-a) + f_y(\vec{a})(y-b) + f_z(\vec{a})(z-c) + f(\vec{a}) \\ &= \big(f_x(\vec{a}), f_y(\vec{a}), f_z(\vec{a})\big) \cdot (x-a, y-b, z-c) \\ &= \boldsymbol{\nabla} f(\vec{a}) \cdot (\vec{x} - \vec{a}) + f(\vec{a}) \end{aligned}$$

📘 **Definition — Gradient Vector**

If the partial derivatives of $f(x,y,z)$ are defined at $\vec{a} \in \mathbb{R}^3$,

then the **gradeient vector** of f at $\vec{a}$ is

$$\boldsymbol{\nabla} f(\vec{a}) = \big(f_x(\vec{a}), f_y(\vec{a}), f_z(\vec{a})\big)$$

Example:

Let $f(x,y,z) = x^2 + z^3 \sin y$. Find the gradient vector of f and the linearization of f at $\vec{a} = (2,\pi,1)$

$$\begin{aligned} L_{\vec{a}}(\vec{x}) \approx \boldsymbol{\nabla} f(\vec{a}) \cdot (\vec{x} - \vec{a}) + f(\vec{a}) &= (4,-1,0) \cdot (x-2, y-\pi, z-1) + 4 \\ &= 4(x-2) - 1(y-\pi) + 4 = 4x - y - 4 + \pi \end{aligned}$$

📘 **Definition — Generalizaiton to $\mathbb{R}^n$**

Let $\vec{x} = (x_1, x_2, ..., x_n) \in \mathbb{R}^n$ and $\vec{a} \in \mathbb{R}^n$

$\vec{x} - \vec{a} = (x_1 - a_1, x_2 - a_2, ..., x_n - a_n)$ and define $\boldsymbol{\nabla} f(\vec{a}) = (D_1 f(\vec{a}_1), D_2 f(\vec{a}_2), ..., D_n f(\vec{a}_n))$

Then

$$f(\vec{x}) \approx f(\vec{a}) = \boldsymbol{\nabla} f(\vec{a})(\vec{x} - \vec{a})$$